ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P1HR Worked Solutions

Jun 2022 | P1HR

31 questions ▪ question image followed by worked solution

Prepared by

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Student Edition

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 4

TOPIC: **Probability Toolkit**

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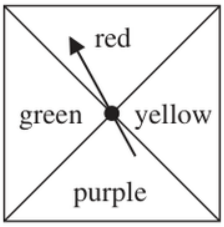
1 Here is a biased spinner.



When the spinner is spun once, the probabilities that it lands on red or on yellow or on green are given in the table.

Colour	red	yellow	purple	green
Probability	0.25	0.2		0.2

(a) Work out the probability that the spinner lands on red or on yellow.

....

(1)

Yang is going to spin the spinner 300 times.

(b) Work out an estimate for the number of times the spinner will land on purple.

....

(3)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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1. Calculate probability

$$P(\text{red or yellow}) = 0.25 + 0.20 = 0.45$$

2. The purple probability is

The purple probability is

$$1 - 0.25 - 0.20 - 0.20 = 0.35$$

3. Expected purple spins

Expected purple spins:

$$300 \times 0.35 = 105$$

FINAL ANSWER

0.45, 0.35, 105.

PAST PAPER QUESTION

Topic: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

Q#: 2 Marks: 4

2 In a warehouse there are two types of shelves, type **R** and type **S**.

These two types of shelves are arranged into shelving units that form a sequence of patterns.

Here are the first three terms in the sequence.

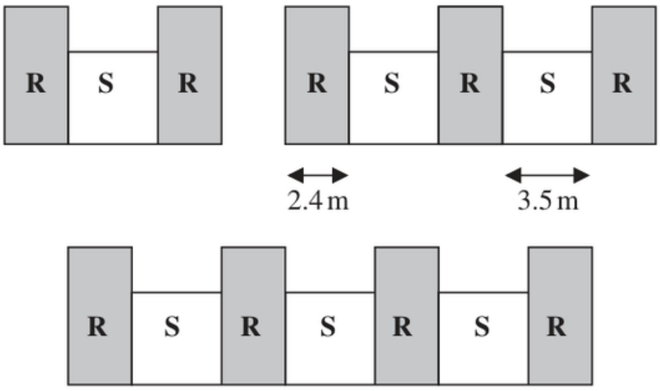

Diagram **NOT** accurately drawn

The width of each type **R** shelf is 2.4 m and the width of each type **S** shelf is 3.5 m

(a) Work out the total width of a shelving unit that has 6 type **R** shelves.

.... m

(2)

A shelving unit has n type **R** shelves.
The total width of this shelving unit is W metres.

(b) Find an expression for W in terms of n
Give your answer in its simplest form.

W = ...

(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Sequences**

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WORKED SOLUTION

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1. Find the nth term

For a shelving unit with 6 type R shelves, there are 5 type S shelves.

$$6(2.4) + 5(3.5) = 14.4 + 17.5 = 31.9$$

2. Find the nth term

For n type R shelves, there are $n - 1$ type S shelves.

$$W = 2.4n + 3.5(n - 1)$$

$$W = 2.4n + 3.5n - 3.5 = 5.9n - 3.5$$

FINAL ANSWER

31.9 m, and $W = 5.9n - 3.5$.

PAST PAPER QUESTION Q#: 3 Marks: 3

TOPIC: **Statistics Toolkit**

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3 Here are five cards.

Each card has a number written on it.

15	7	–2	23	x

The mean of the five numbers is 12

Work out the value of x

$x =$ ....

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

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1. The mean of the five

The mean of the five cards is 12, so the total is

$$5 \times 12 = 60$$

2. The known total is

The known total is

$$15 + 7 - 2 + 23 = 43$$

$$x = 60 - 43 = 17$$

FINAL ANSWER

$x = 17$.

PAST PAPER QUESTION

Topic: **Percentages**

Jun 2022 P1HR - Jun 2022 | P1HR

Q#: 4 Marks: 4

4 The language department of a college has 180 students.
Each student studies exactly one of French, German, Italian or Spanish.

15 students study French.
45% of the students study German.

Express the percentage of students studying Italian or Spanish as a percentage of those studying French or German.

.... %

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Percentages**

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WORKED SOLUTION

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1. Total students

Total students:

$$180$$

2. French

French:

$$15$$

3. German

German:

$$45\% \times 180 = 81$$

4. French or German

French or German:

$$15 + 81 = 96$$

5. Italian or Spanish

Italian or Spanish:

$$180 - 96 = 84$$

6. Convert standard form

Write 84 as a percentage of 96:

$$\frac{84}{96} \times 100 = 87.5$$

FINAL ANSWER

87.5%

PAST PAPER QUESTION Q#: 5 Marks: 5

TOPIC: **Factorising**

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5 (a) Expand $3c^3(c + 4)$

....
(2)

(b) (i) Factorise $x^2 + 8x - 9$

....
(2)

(ii) Hence, solve $x^2 + 8x - 9 = 0$

....
(1)

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 5

TOPIC: **Factorising**

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WORKED SOLUTION

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1. (a)

(a) $3c^3(c+4) = 3c^4 + 12c^3$.

2. (b)(i)

(b)(i)

$$x^2 + 8x - 9 = (x+9)(x-1)$$

3. (b)(ii)

(b)(ii)

$$(x+9)(x-1) = 0$$

FINAL ANSWER

(a) $3c^4 + 12c^3$, (b)(i) $(x+9)(x-1)$, (b)(ii) $x = -9$ or $x = 1$

PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Fractions**

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6 Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Fractions**

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WORKED SOLUTION

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1. Simplify fraction

$$2\frac{2}{3} = \frac{8}{3}$$

$$3\frac{3}{4} = \frac{15}{4}$$

2. Use a common denominator

Use a common denominator:

$$\frac{8}{3} + \frac{15}{4} = \frac{32}{12} + \frac{45}{12}$$

$$= \frac{77}{12} = 6\frac{5}{12}$$

FINAL ANSWER

$6\frac{5}{12}$

PAST PAPER QUESTION

Q#: 7 Marks: 3

TOPIC: **Standard & Compound Units**

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7 The diagram shows a solid cylinder made from iron.

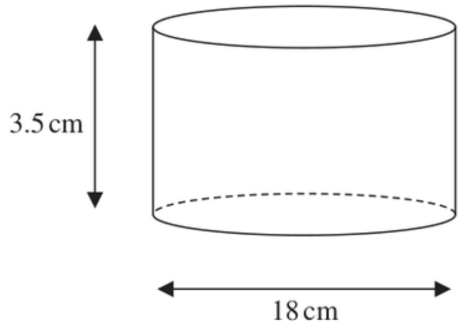

Diagram **NOT** accurately drawn

The cylinder has diameter 18 cm and height 3.5 cm
The mass of the cylinder is 7.04 kg

Work out the density of the iron.
Give your answer in g/cm^3 correct to 2 significant figures.

.... g/cm^3

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Standard & Compound Units**

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WORKED SOLUTION

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1. Radius of the cylinder is

Radius of the cylinder is

$$\frac{18}{2} = 9 \text{ cm}$$

2. Volume is

Volume is

$$\pi r^2 h = \pi(9^2)(3.5) = 283.5\pi$$

$$= 890.64\ldots \text{ cm}^3$$

3. Mass is

Mass is

$$7.04 \text{ kg} = 7040 \text{ g}$$

4. Density is

Density is

$$\frac{7040}{890.64\ldots} = 7.904\ldots$$

FINAL ANSWER

7.9 g/cm^3 to 2 significant figures.

PAST PAPER QUESTION

Topic: **Compound Interest & Depreciation**

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Q#: 8 Marks: 3

8 Jane bought a new car for \$18 000
The car depreciates in value by 15% each year.

Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest \$

\$..

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Compound Interest & Depreciation**

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WORKED SOLUTION

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1. A depreciation of gives multiplier

A depreciation of 15% gives multiplier

$$0.85$$

2. Use compound interest

After 4 years,

$$18000(0.85)^4 = 9396.1125$$

3. Use compound interest

Correct to the nearest dollar,

$$9396.1125 \approx 9396$$

FINAL ANSWER

$9396

PAST PAPER QUESTION

Q#: 9 Marks: 2

TOPIC: **Solving Inequalities**

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9 Solve the inequality $3 - 4x \leqslant 11$

(Total for Question 9 is 2 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 2

TOPIC: **Solving Inequalities**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

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1. Solve inequality

$$3 - 4x \leq 11$$

$$-4x \leq 8$$

2. Solve inequality

Divide by -4, reversing the inequality:

$$x \geq -2$$

FINAL ANSWER

$x \geq -2$.

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Linear Graphs ($y = mx + c$)**

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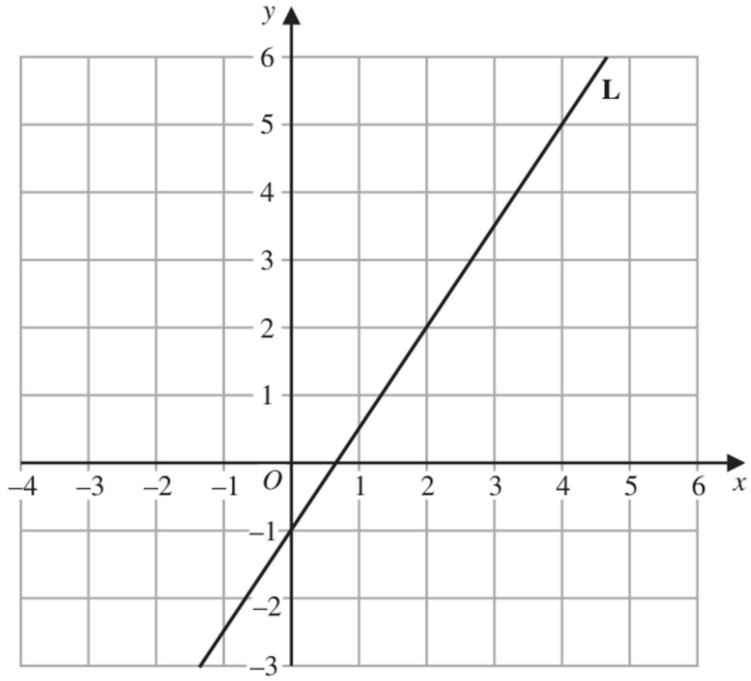
10 Line **L** is drawn on the grid.



Find an equation for **L**
Give your answer in the form $y = mx + c$

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Linear Graphs (y = mx + c)**

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WORKED SOLUTION

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1. Solve equation

From the graph, the line crosses the y-axis at -1, so

$$c = -1$$

2. Find the gradient

Using two points on the line, for example $(0, -1)$ and $(2, 2)$, the gradient is

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

3. Solve equation

So the equation is

$$y = \frac{3}{2}x - 1$$

FINAL ANSWER

$y = \frac{3}{2}x - 1.$

PAST PAPER QUESTION Q#: 11 Marks: 6

TOPIC: **Area & Perimeter**

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11 The diagram shows a quadrilateral *ABCD*

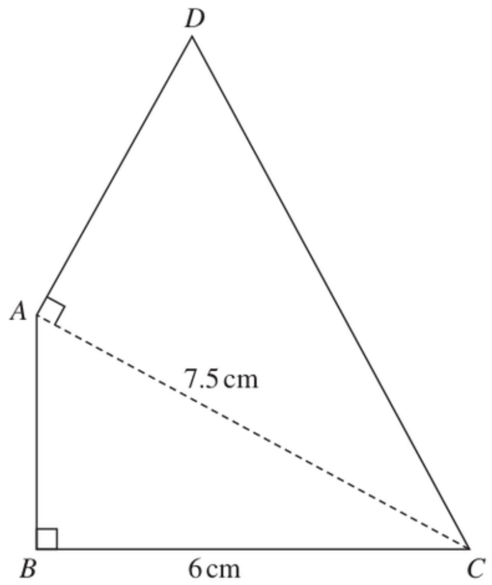

Diagram **NOT** accurately drawn

In the diagram, *ABC* and *DAC* are right-angled triangles.

$BC = 6\,\text{cm}$ $AC = 7.5\,\text{cm}$

The area of quadrilateral *ABCD* is $31.5\,\text{cm}^2$

Work out the length of *AD*

Question 11 continued.

.... cm

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 6

TOPIC: **Area & Perimeter**

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WORKED SOLUTION

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1. Use trigonometry

In right-angled triangle ABC,

$$AB^2 + BC^2 = AC^2$$

$$AB^2 + 6^2 = 7.5^2$$

$$AB^2 = 56.25 - 36 = 20.25$$

$$AB = 4.5 \text{ cm}$$

2. Area of triangle

Area of triangle ABC:

$$\frac{1}{2} \times 6 \times 4.5 = 13.5 \text{ cm}^2$$

3. Area of triangle

Area of triangle ADC:

$$31.5 - 13.5 = 18 \text{ cm}^2$$

4. Use trigonometry

Triangle ADC is right-angled, so

$$\frac{1}{2} \times AD \times 7.5 = 18$$

$$AD = \frac{36}{7.5} = 4.8$$

FINAL ANSWER

4.8 cm.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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12 $P = 3^3 \times 5^2 \times 7$
$Q = 3^2 \times 5 \times 7^2$

(a) Write down the highest common factor (HCF) of P and Q

....
(1)

$P = 3^3 \times 5^2 \times 7$
$Q = 3^2 \times 5 \times 7^2$

(b) Work out the value of $P^3 \times Q$
Give your answer in the form $3^x \times 5^y \times 7^z$ where x, y and z are positive integers.

....
(2)

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION Q#: 12 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

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1. Find the LCM

$$P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

2. Find the LCM

For the HCF, take the smaller power of each common prime:

$$\text{HCF} = 3^2 \times 5 \times 7$$

3. Find the LCM

For $P^3 \times Q$:

$$P^3 = (3^3 \times 5^2 \times 7)^3 = 3^9 \times 5^6 \times 7^3$$

4. Find the LCM

Now multiply by Q:

$$P^3 \times Q = (3^9 \times 5^6 \times 7^3)(3^2 \times 5 \times 7^2)$$

$$P^3 \times Q = 3^{11} \times 5^7 \times 7^5$$

FINAL ANSWER

(a) $3^2 \times 5 \times 7$, (b) $3^{11} \times 5^7 \times 7^5$

PAST PAPER QUESTION

Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

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13 Here is the number of runs scored by a baseball team in each of its 15 games this season.

The number of runs have been arranged in order of size.

0 1 1 3 5 6 7 7 8 9 9 12 12 15 16

Work out the interquartile range of the number of runs.

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 2

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

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1. Use Statistics Toolkit correct

The runs are already in order:

$$0,\ 1,\ 1,\ 3,\ 5,\ 6,\ 7,\ 7,\ 8,\ 9,\ 9,\ 12,\ 12,\ 15,\ 16$$

2. Calculate statistic

There are 15 values, so ignore the median when finding the quartiles.

$$Q_1 = 3, \qquad Q_3 = 12$$

$$\text{IQR} = 12 - 3 = 9$$

FINAL ANSWER

9.

PAST PAPER QUESTION

Topic: **Simultaneous Equations**

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Q#: 14 Marks: 4

14 Solve the simultaneous equations

$$3x - 5y = 25$$
$$4x + 3y = 14$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Simultaneous Equations**

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WORKED SOLUTION

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1. Solve simultaneous equations

$$3x - 5y = 25$$

$$4x + 3y = 14$$

2. Multiply the first equation by

Multiply the first equation by 3:

$$9x - 15y = 75$$

3. Multiply the second equation by

Multiply the second equation by 5:

$$20x + 15y = 70$$

4. Add

Add:

$$29x = 145$$

$$x = 5$$

5. Solve simultaneous equations

Substitute into $3x - 5y = 25$:

$$15 - 5y = 25$$

$$-5y = 10$$

$$y = -2$$

FINAL ANSWER

$x = 5$, $y = -2$.

PAST PAPER QUESTION

Q#: 15 Marks: 3

TOPIC: **Circle Theorems**

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15

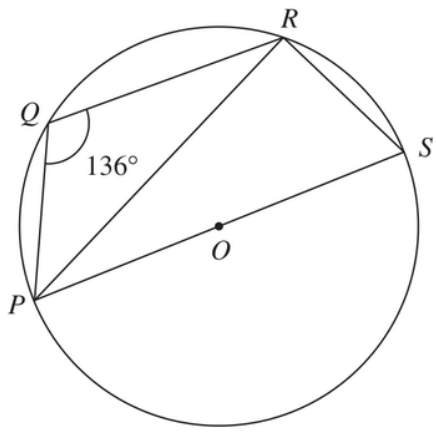

Diagram **NOT** accurately drawn

P, *Q*, *R* and *S* are points on a circle with centre *O*

PS is a diameter of the circle.

Angle $PQR = 136°$

Work out the size of angle *RPS*

.... °

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION Q#: 15 Marks: 3

TOPIC: **Circle Theorems**

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WORKED SOLUTION

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1. Opposite angles in a cyclic

Opposite angles in a cyclic quadrilateral add to 180°:

$$\angle PSR = 180° - 136° = 44°$$

2. Use trigonometry

Since PS is a diameter,

$$\angle PRS = 90°$$

3. Use trigonometry

In triangle PRS,

$$\angle RPS = 180° - 90° - 44° = 46°$$

FINAL ANSWER

46°.

PAST PAPER QUESTION Q#: 16 Marks: 6

TOPIC: **Expanding Brackets**

Jun 2022 P1HR - Jun 2022 | P1HR

16 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$

....

(3)

(b) Simplify fully $\left(\dfrac{2x^5}{8xy^2}\right)^{-2}$

....

(3)

(Total for Question 16 is 6 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 6

TOPIC: **Expanding Brackets**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

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1. (a)

(a)

$$(3x-1)(x+2)(3x+1) = (9x^2-1)(x+2)$$

$$= 9x^3 + 18x^2 - x - 2$$

2. (b)

(b)

$$\left(\frac{2x^5}{8xy^2}\right)^{-2} = \left(\frac{x^4}{4y^2}\right)^{-2}$$

$$= \left(\frac{4y^2}{x^4}\right)^2 = \frac{16y^4}{x^8}$$

FINAL ANSWER

(a) $9x^3 + 18x^2 - x - 2$, (b) $\frac{16y^4}{x^8}$

PAST PAPER QUESTION

Q#: 17 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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17 Here is a parallelogram *PQRS*, in which angle *SPQ* is acute.

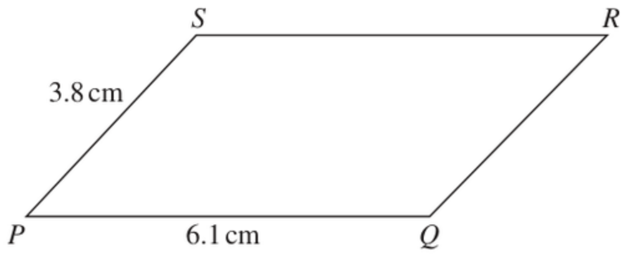

Diagram **NOT** accurately drawn

$PQ = 6.1$ cm $\quad\quad PS = 3.8$ cm

The area of the parallelogram is 18 cm^2

Work out the length of *QS*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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WORKED SOLUTION

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1. Height of the parallelogram

Height of the parallelogram:

$$h = \frac{18}{6.1} = 2.9508\ldots$$

2. Use cosine rule

The horizontal offset from P to S is

$$\sqrt{3.8^2 - h^2} = \sqrt{3.8^2 - 2.9508\ldots^2} = 2.394\ldots$$

3. Use trigonometry

So the horizontal distance from S to Q is

$$6.1 - 2.394\ldots = 3.705\ldots$$

$$QS = \sqrt{3.705\ldots^2 + 2.9508\ldots^2} = 4.737\ldots$$

FINAL ANSWER

4.74 cm.

PAST PAPER QUESTION Q#: 18 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2022 P1HR - Jun 2022 | P1HR

18 The diagram shows a cube *ABCDEFGH* with sides of length 6 cm.

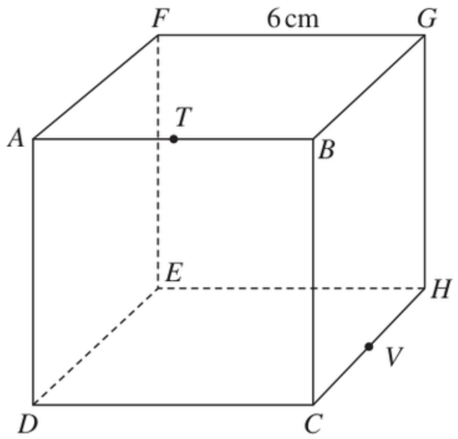

Diagram **NOT** accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.
Give your answer in the form $\sqrt{a}$ cm where a is an integer.

.... cm

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: 3D Pythagoras & Trigonometry

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WORKED SOLUTION

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1. The cube has side length

The cube has side length 6 cm.

2. Find the midpoint

Let T be the midpoint of AB and V be the midpoint of CH.

3. The changes in the three

The changes in the three perpendicular directions from T to V are

$$3, \quad 3, \quad 6$$

4. Use trigonometry

So by 3D Pythagoras,

$$TV = \sqrt{3^2 + 3^2 + 6^2}$$

$$TV = \sqrt{9 + 9 + 36} = \sqrt{54}$$

FINAL ANSWER

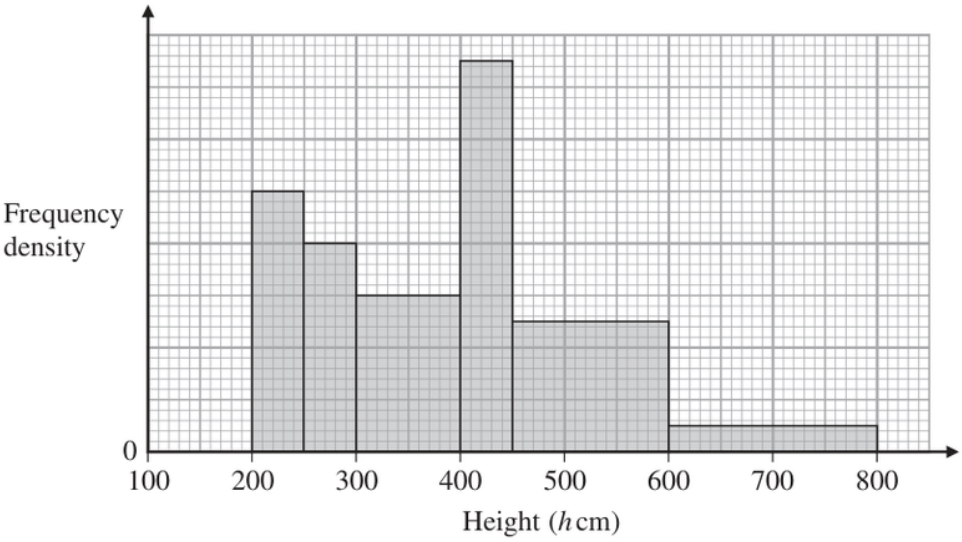
$\sqrt{54}$ cm.

PAST PAPER QUESTION Q#: 19 Marks: 3

TOPIC: **Histograms**

Jun 2022 P1HR - Jun 2022 | P1HR

19 The histogram gives information about the height, h cm, of each tree in part of a forest.



There are no trees for which $h \leqslant 200$ and for which $h > 800$

The number of trees for which $300 < h \leqslant 400$ is 8 fewer than the number of trees for which $400 < h \leqslant 500$

Work out an estimate for the number of trees in this part of the forest that have a height greater than 500 cm.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION Q#: 19 Marks: 3

TOPIC: **Histograms**

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WORKED SOLUTION

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1. Use relative densities from the

Use relative densities from the histogram:

$$10,\ 8,\ 6,\ 15,\ 5,\ 1$$

2. Use histogram

The number of trees with $300 < h \leq 400$ is

$$100(6s) = 600s$$

3. Use histogram

The number of trees with $400 < h \leq 500$ is

$$50(15s) + 50(5s) = 1000s$$

4. The difference is

The difference is 8:

$$1000s - 600s = 8$$

$$400s = 8$$

$$s = 0.02$$

5. Use histogram

For height greater than 500:

$$100(5s) + 200(1s) = 700s$$

$$700(0.02) = 14$$

FINAL ANSWER

14 trees.

PAST PAPER QUESTION

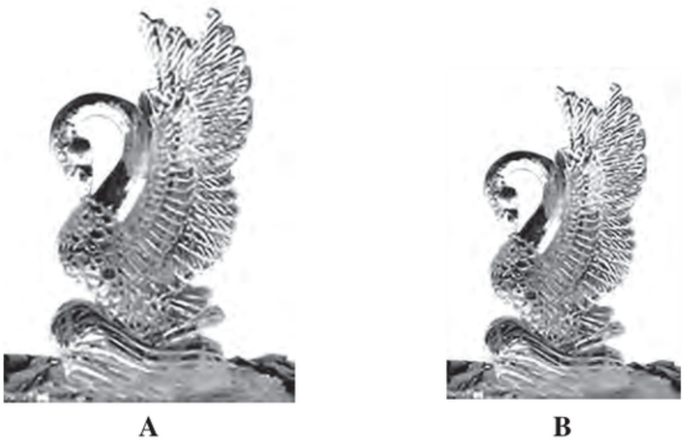
Q#: 20 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

20 The diagram shows two similar metal statues.

Diagram **NOT** accurately drawn



The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is k% less than the surface area of statue **A**

Work out the value of k
Give your answer correct to 3 significant figures.

k = ...

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

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WORKED SOLUTION

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1. The volume of statue is

The volume of statue B is 20% less than statue A, so the volume scale factor is 0.8.

2. Calculate area

For similar solids, the linear scale factor is the cube root of the volume scale factor:
linear scale factor $= \sqrt[3]{0.8}$

3. Surface area scale factor is

Surface area scale factor is the square of the linear scale factor:
surface area scale factor $= (\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773...$

4. Calculate area

So statue B has 86.1773...% of the surface area of statue A.
$k = 100 - 86.1773... = 13.8226...$

FINAL ANSWER
$k = 13.8$ to 3 significant figures.

PAST PAPER QUESTION

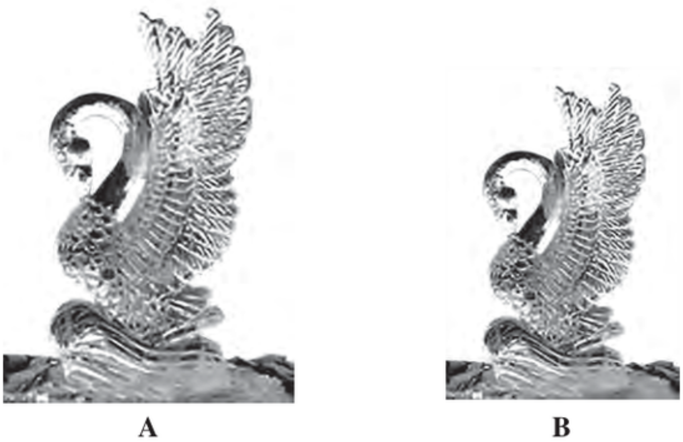
Topic: **Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

Q#: 20 Marks: 4

20 The diagram shows two similar metal statues.

Diagram **NOT** accurately drawn



The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is k% less than the surface area of statue **A**

Work out the value of k
Give your answer correct to 3 significant figures.

k = ...

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The volume of statue is

The volume of statue B is 20% less than statue A, so the volume scale factor is 0.8.

2. Calculate area

For similar solids, the linear scale factor is the cube root of the volume scale factor:
linear scale factor $= \sqrt[3]{0.8}$

3. Surface area scale factor is

Surface area scale factor is the square of the linear scale factor:
surface area scale factor $= (\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773...$

4. Calculate area

So statue B has 86.1773...% of the surface area of statue A.
$k = 100 - 86.1773... = 13.8226...$

FINAL ANSWER
$k = 13.8$ to 3 significant figures.

PAST PAPER QUESTION

Q#: 21 Marks: 4

TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

21 Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify surd

$$\frac{3+\sqrt{8}}{(\sqrt{2}-1)^2}$$

2. Simplify surd

Since $\sqrt{8}=2\sqrt{2}$,

$$\frac{3+2\sqrt{2}}{(\sqrt{2}-1)^2}$$

3. Simplify surd

Also

$$(\sqrt{2}-1)^2 = 3-2\sqrt{2}$$

4. Simplify surd

So

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}} \times \frac{3+2\sqrt{2}}{3+2\sqrt{2}} = \frac{(3+2\sqrt{2})^2}{9-8}$$

$$= (3+2\sqrt{2})^2 = 9+12\sqrt{2}+8$$

$$= 17+12\sqrt{2} = 17+\sqrt{288}$$

FINAL ANSWER

$17+\sqrt{288}$

PAST PAPER QUESTION

Q#: 21 **Marks: 4**

TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

21 Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify surd

$$\frac{3+\sqrt{8}}{(\sqrt{2}-1)^2}$$

2. Simplify surd

First simplify:

$$\sqrt{8} = 2\sqrt{2}$$

3. Simplify surd

and

$$(\sqrt{2}-1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$

4. Simplify surd

So

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}}$$

5. Rationalise the denominator

Rationalise the denominator:

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}} \times \frac{3+2\sqrt{2}}{3+2\sqrt{2}} = \frac{(3+2\sqrt{2})^2}{9-8}$$

$$(3+2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

FINAL ANSWER

$17 + 12\sqrt{2}$

PAST PAPER QUESTION

Q#: 22 Marks: 7

TOPIC: **Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

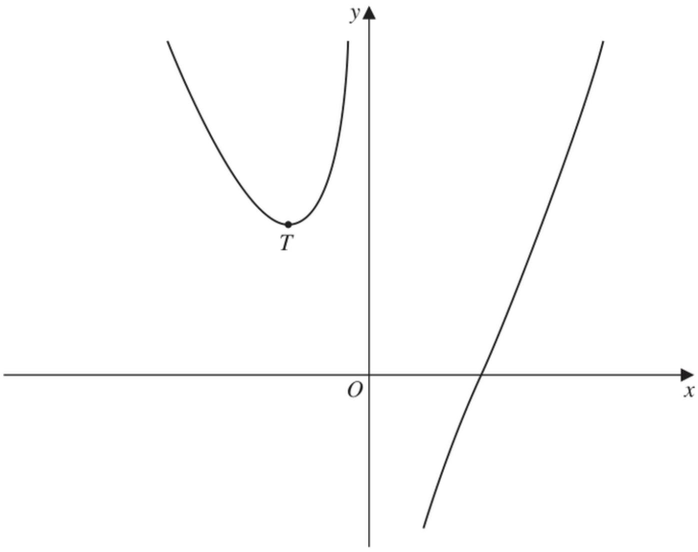


Diagram **NOT** accurately drawn

For all values of p, the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

(a) find the value of p

$p =$ ....

(4)

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

(b) Find the value of k

$k =$ ....

(3)

(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 7

TOPIC: **Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find the gradient

For part (a),

$$y = x^2 - \frac{p}{x}$$

$$\frac{dy}{dx} = 2x + \frac{p}{x^2}$$

2. Find the gradient

At the minimum point, $x = -3$ and the gradient is 0.

$$2(-3) + \frac{p}{(-3)^2} = 0$$

$$-6 + \frac{p}{9} = 0$$

$$p = 54$$

3. Find the gradient

For part (b),

$$y = x^2 - \frac{16}{x}$$

$$\frac{dy}{dx} = 2x + \frac{16}{x^2}$$

4. Find the gradient

At a horizontal tangent,

$$2x + \frac{16}{x^2} = 0$$

$$2x^3 + 16 = 0$$

$$x^3 = -8$$

$$x = -2$$

5. Find the gradient

Now substitute $x = -2$ into the curve:

$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

FINAL ANSWER
$p = 54,\ k = 12$

PAST PAPER QUESTION

Q#: 22 Marks: 7

TOPIC: **Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

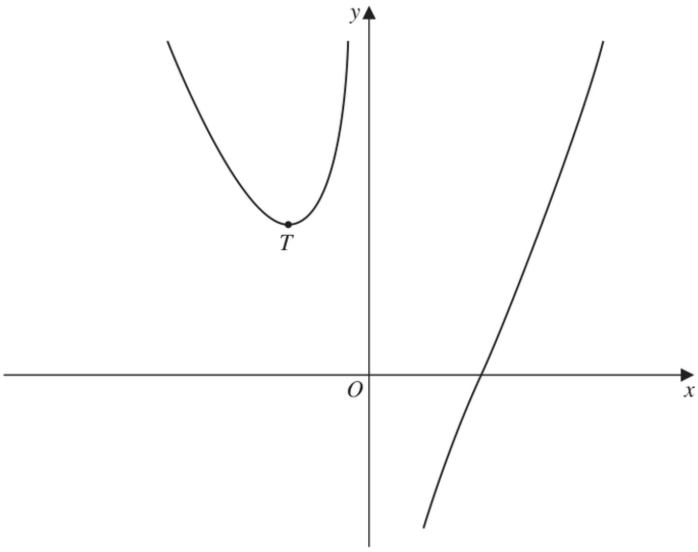


Diagram **NOT** accurately drawn

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$p =$ ....

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(b) Find the value of k

$k =$ ....

(3)

(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 7

Topic: **Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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$$x = -2$$

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Now substitute $x = -2$ into the curve:

$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

FINAL ANSWER
$p = 54,\ k = 12$

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

....
(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.......................... , ...)
(2)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a)

$$2x^2 - 12x + 3 = 2(x^2 - 6x) + 3$$

$$= 2\left((x-3)^2 - 9\right) + 3$$

$$= 2(x-3)^2 - 18 + 3$$

$$= 2(x-3)^2 - 15$$

2. (b)

(b)

$$y = 2(x+4)^2 - 12(x+4) + 3$$

3. Let

Let $u = x + 4$. From part (a),

$$2u^2 - 12u + 3 = 2(u-3)^2 - 15$$

4. Calculate value

So

$$y = 2((x+4) - 3)^2 - 15$$

$$y = 2(x+1)^2 - 15$$

5. The minimum point is when

The minimum point is when $x = -1$, giving $y = -15$.

FINAL ANSWER

(a) $2(x-3)^2 - 15$, (b) $M = (-1, -15)$

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

....

(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.......................... , ...)

(2)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a)

$$2x^2 - 12x + 3 = 2(x^2 - 6x) + 3$$

$$= 2\left((x-3)^2 - 9\right) + 3$$

$$= 2(x-3)^2 - 18 + 3$$

$$= 2(x-3)^2 - 15$$

2. (b)

(b)

$$y = 2(x+4)^2 - 12(x+4) + 3$$

3. Let

Let $u = x + 4$. From part (a),

$$2u^2 - 12u + 3 = 2(u-3)^2 - 15$$

4. Calculate value

So

$$y = 2((x+4) - 3)^2 - 15$$

$$y = 2(x+1)^2 - 15$$

5. The minimum point is when

The minimum point is when $x = -1$, giving $y = -15$.

FINAL ANSWER

(a) $2(x-3)^2 - 15$, (b) $M = (-1, -15)$

PAST PAPER QUESTION

TOPIC: **Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

Q#: 24 Marks: 5

24 Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.

The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of x
Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. There are counters. Initially 5

There are x counters. Initially 5 show red, so $x - 5$ show green.

2. List probability cases

After two turns, there are still 5 red faces if:

3. the same counter is picked

- the same counter is picked twice, or
- one red counter and one green counter are picked.

4. The probability of picking the

The probability of picking the same counter twice is

$$\frac{1}{x}$$

5. The probability of picking one

The probability of picking one red and one green in either order is

$$\frac{5}{x} \cdot \frac{x-5}{x} + \frac{x-5}{x} \cdot \frac{5}{x}$$

$$= \frac{10(x-5)}{x^2}$$

6. Evaluate fraction

So

$$\frac{1}{x} + \frac{10(x-5)}{x^2} = \frac{19}{32}$$

$$\frac{11x - 50}{x^2} = \frac{19}{32}$$

$$32(11x - 50) = 19x^2$$

$$19x^2 - 352x + 1600 = 0$$

$$(x - 8)(19x - 200) = 0$$

7. Calculate value

Since x must be a whole number,

$$x = 8$$

FINAL ANSWER

$x = 8$

PAST PAPER QUESTION

Q#: 24 **Marks: 5**

TOPIC: **Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

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The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of x
Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. There are counters. Initially 5

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7. Calculate value

Since x must be a whole number,

$$x = 8$$

FINAL ANSWER
$x = 8$

PAST PAPER QUESTION Q#: 25 Marks: 5

TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

25 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$S_{10} = 4S_5$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$S_5 = \frac{5}{2}(2a + 4d) = 5(a + 2d)$$

3. Calculate value

So

$$5(2a + 9d) = 4 \cdot 5(a + 2d)$$

$$2a + 9d = 4a + 8d$$

$$d = 2a$$

4. The 8th term is 45

The 8th term is 45:

$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

The first term is 3

PAST PAPER QUESTION

Q#: 25 Marks: 5

TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

25 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate value

$$S_{10} = 4S_5$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$S_5 = \frac{5}{2}(2a + 4d) = 5(a + 2d)$$

3. Calculate value

So

$$5(2a + 9d) = 4 \cdot 5(a + 2d)$$

$$2a + 9d = 4a + 8d$$

$$d = 2a$$

4. The 8th term is 45

The 8th term is 45:

$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

The first term is 3